BODY CONTROL MODULE (BCM)

id094000004100

Purpose

- The BCM performs control of multiple systems such as the power door locks and the windshield wipers and washer based on the input signals such as from the light switch, front door key cylinder switch, and the wiper and washer switch.

Function

- The BCM control content is as follows.

Control table

Control	Content	Reference
Front fog light	The BCM performs control of the front fog light relay based on the signal from the front fog light switch.	(See FRONT FOG LIGHT.)
Turn light system	The BCM performs control of the turn light system based on the signal from the turn switch.	(See TURN LIGHT SYSTEM.)
Brake light	The BCM performs control of the stop light relay based on the signal from the DSC HU/CM.	(See SMART CITY BRAKE SUPPORT (SCBS).)
Front map light	The BCM performs control of the front map light based on the signals from the door latch switch and interior light switch.	(See FRONT MAP LIGHT.)
Rear map light	The BCM performs control of the rear map light based on the signals from the door latch switch and interior light switch.	(See REAR MAP LIGHT.)
Cargo compartment light	The BCM performs control of the cargo compartment light based on the signals from the liftgate switch and cargo compartment light switch.	(See CARGO COMPARTMENT LIGHT.)
Coming home light system	The BCM performs control of the coming home light system based on the signals from the light switch and the door light switch.	(See COMING HOME LIGHT SYSTEM.)
Leaving home light system	The BCM performs control of the leaving home light system based on the signals from the light switch and the remote transmitter.	(See LEAVING HOME LIGHT SYSTEM.)
Power door lock system	The BCM performs control of the power door lock system based on the signal from the door look link switch or front door key cylinder switch.	(See POWER DOOR LOCK SYSTEM.)
Liftgate opener system	The BCM performs control of the liftgate opener system based on the signal from the liftgate opener switch.	(See LIFTGATE OPENER SYSTEM.)
Windshield wiper	The BCM performs control of the windshield wipers based on the signal from the wiper and washer switch.	(See WINDSHIELD WIPER/ WASHER SYSTEM.)
Rear wiper and washer system	The BCM performs control of the rear wiper and washer system based on the signal from the wiper and washer switch.	(See REAR WIPER/WASHER SYSTEM.)
Blower system	The BCM performs control of the blower relay based on the request signal from the PCM.	(See BLOWER SYSTEM.)
Rear window defogger	The BCM performs control of the rear window defogger relay based on the signal from the climate control unit.	(See REAR WINDOW DEFOGGER/OUTER MIRROR HEATER.)
Horn	The BCM performs control of the horn relay based on the signal from the horn system.	(See HORN.)
Headlight cleaner	The BCM controls the headlight cleaner based on the signal from the wiper and washer switch.	(See HEADLIGHT CLEANER SYSTEM.)
Theft-deterrent system	The BCM controls the theft-deterrent system based on the signals from the door latch switch, liftgate latch switch, hood latch switch, and door lock link switch.	(See THEFT-DETERRENT SYSTEM.)

Battery Discharge Suppression Function

- When the PCM is performing i-ELOOP control and i-stop control, the PCM stops the following functions based on the request signal from the PCM to prevent a vehicle malfunction due to a decrease in power supply voltage.
  - Rear window defogger
  - Blower motor*1
  - PTC heater (With PTC heater)

*1 : If the battery voltage drops to the extent it can be assumed that re-starting the engine will be difficult, cut off the voltage to the blower motor to prioritize re-starting the engine.

Dark current reduction function

- When the ignition is switched OFF (LOCK) or in the ACC position and all of the following conditions are met, the BCM cuts off the power supply input circuit of the CAN communication and BCM to reduce dark current.

— 20 s or more have elapsed since ignition was switched OFF (LOCK) or in ACC position
— Turn light output control is off
— Horn output control is off
— ISO communication between the start stop unit and BCM not performed
— Security indicator light is not turned on/flashing
— LIN communication between the electrical supply unit (ESU) and the BCM is not performed

Automatic configuration function

- When the ignition is switched ON after the BCM is replaced, the BCM reads the vehicle specification information sent via CAN signal from the instrument cluster, and stores the vehicle specification information.

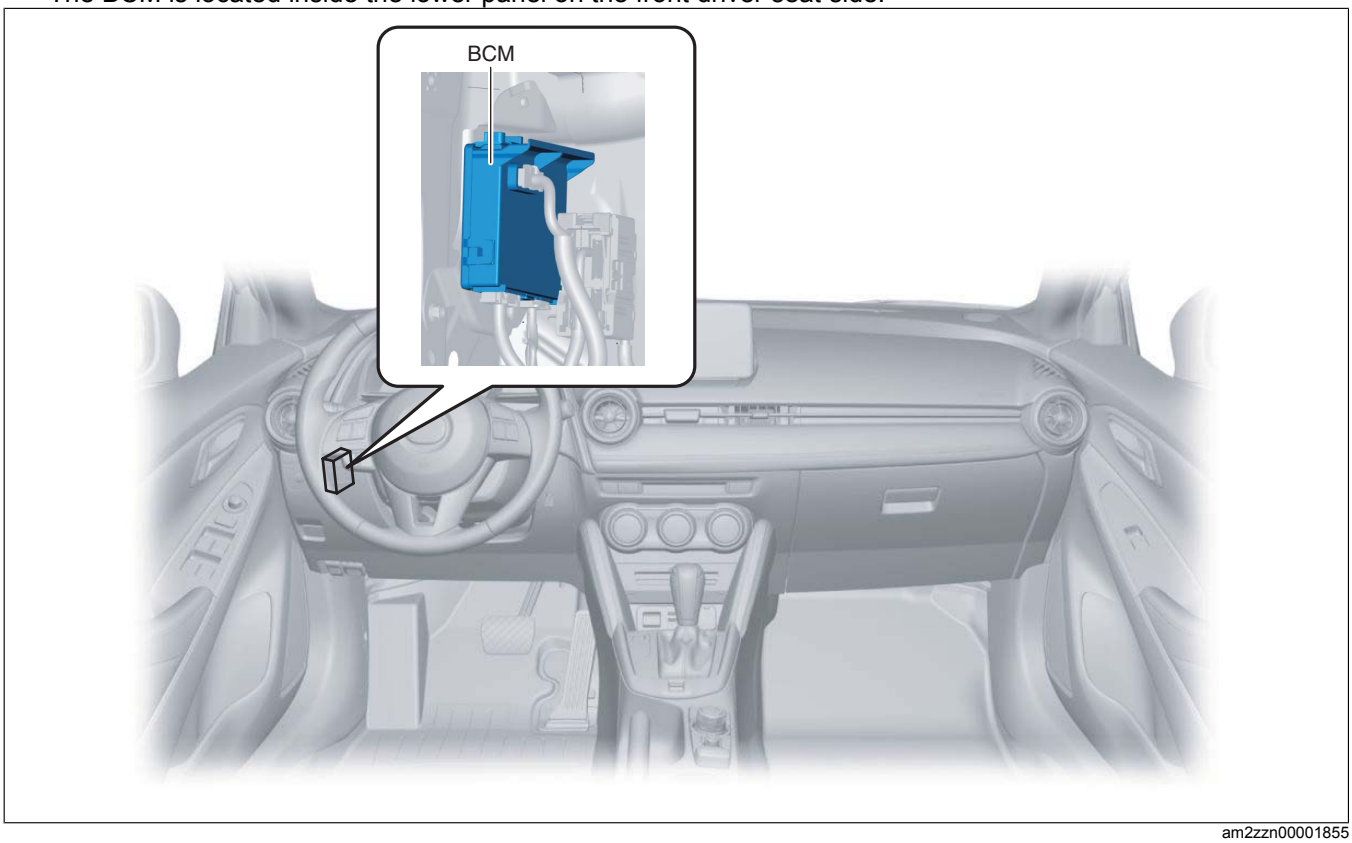
Note

- The personalization setting content stored by the BCM cannot be read by auto configuration. If the personalization setting content is to be read from the pre-replacement BCM to the post-replacement BCM, implement the BCM configuration (use read/write function). For the BCM configuration (using the read/write function) procedure, refer to the BODY CONTROL MODULE (BCM) CONFIGURATION (USING READ/WRITE FUNCTION) in the workshop manual.

Construction

L.H.D.

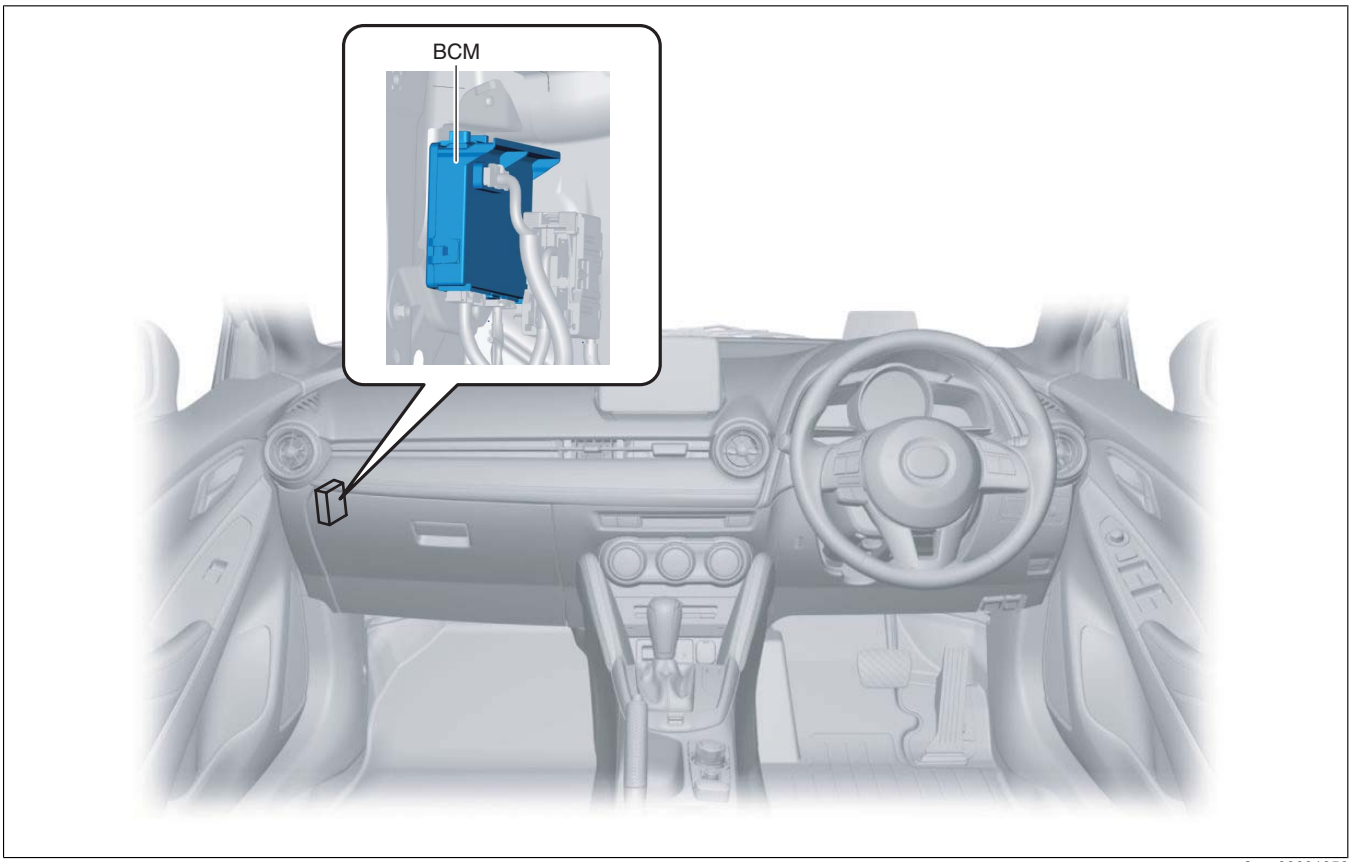
- The BCM is located inside the lower panel on the front driver seat side.



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R.H.D.

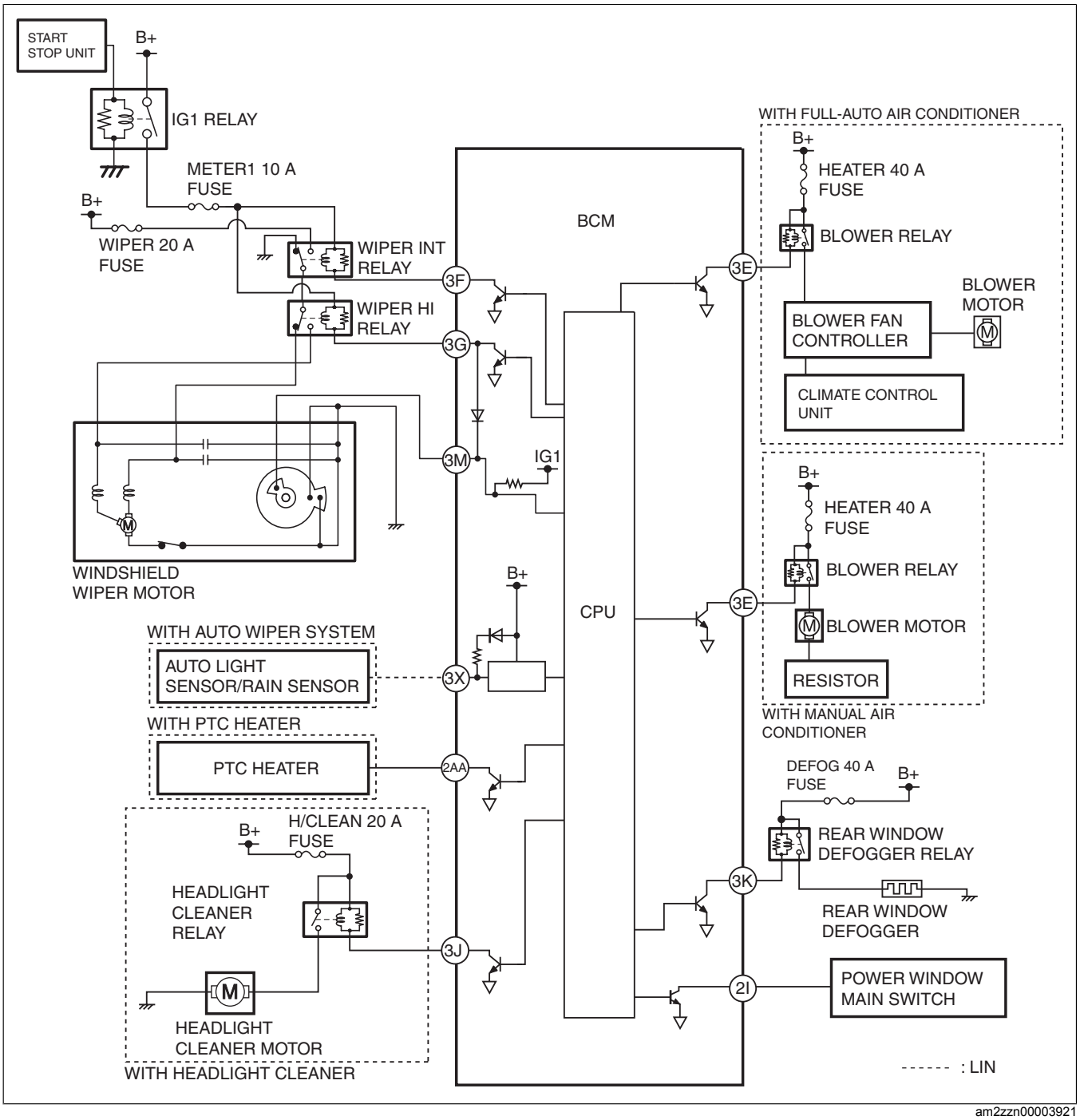
- The BCM is located inside the lower panel on the front passenger seat side.



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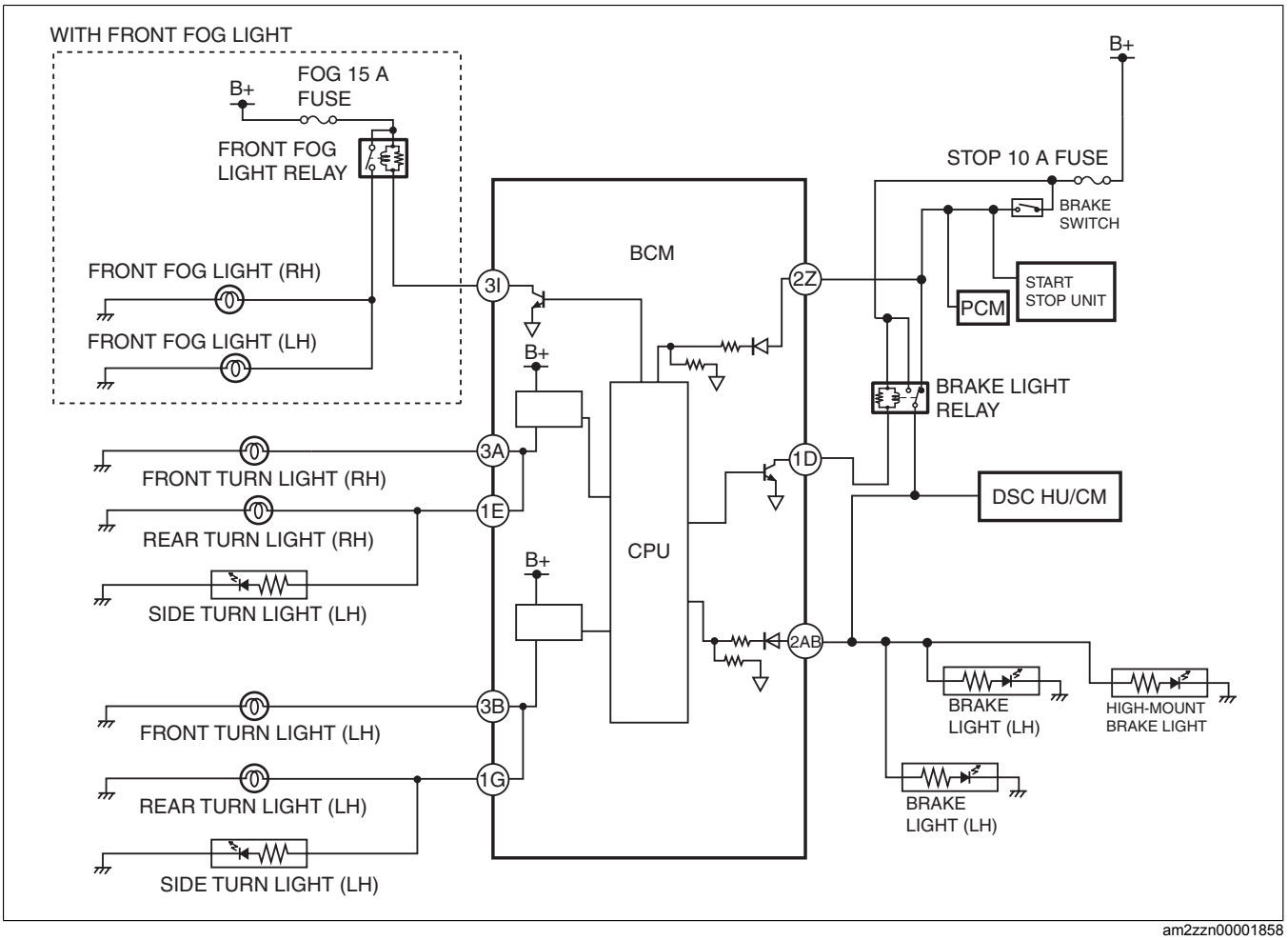
System Wiring Diagram

Windshield wiper, rear window defogger, blower motor, power window, headlight cleaner



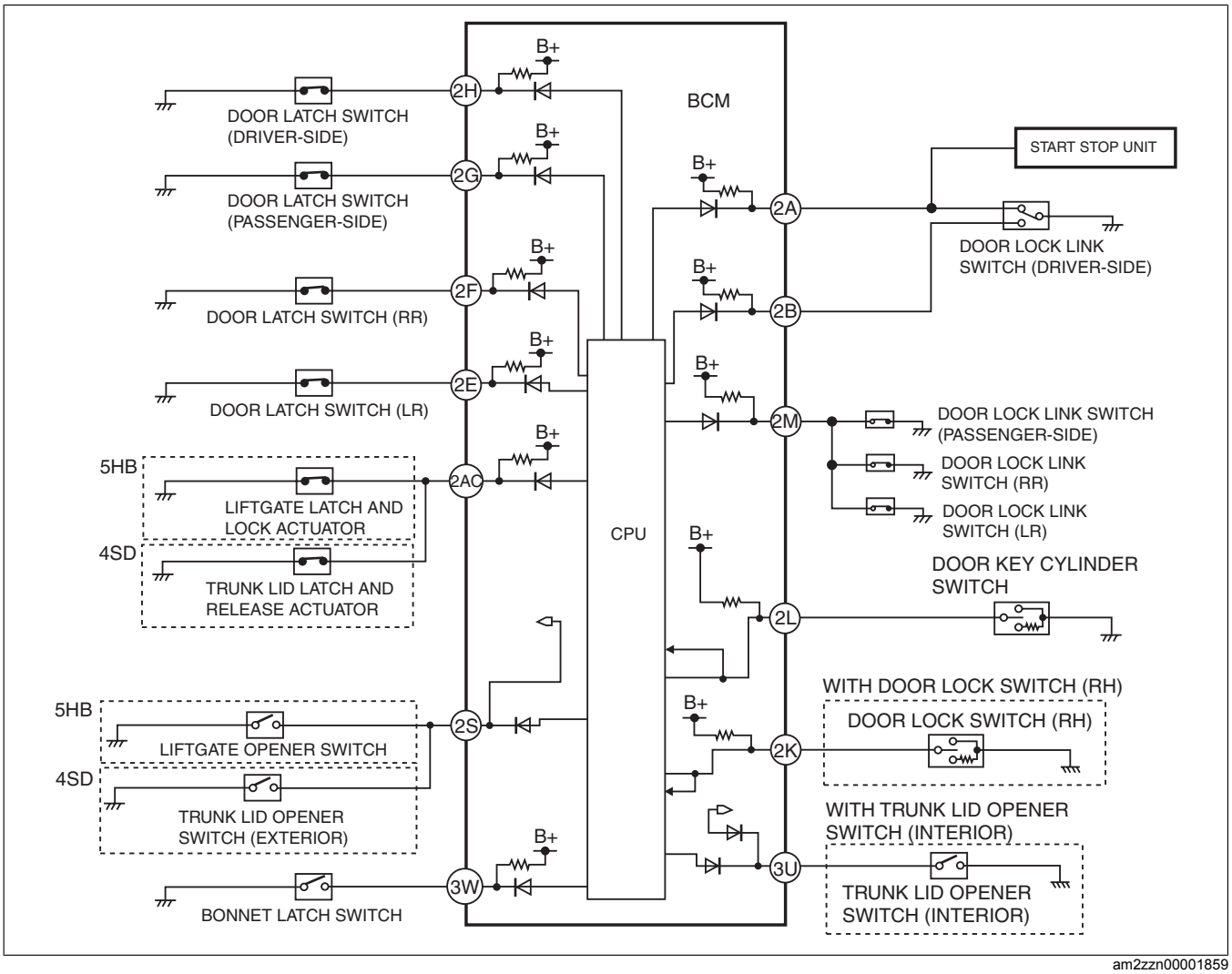
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Front fog light, turn light, brake light



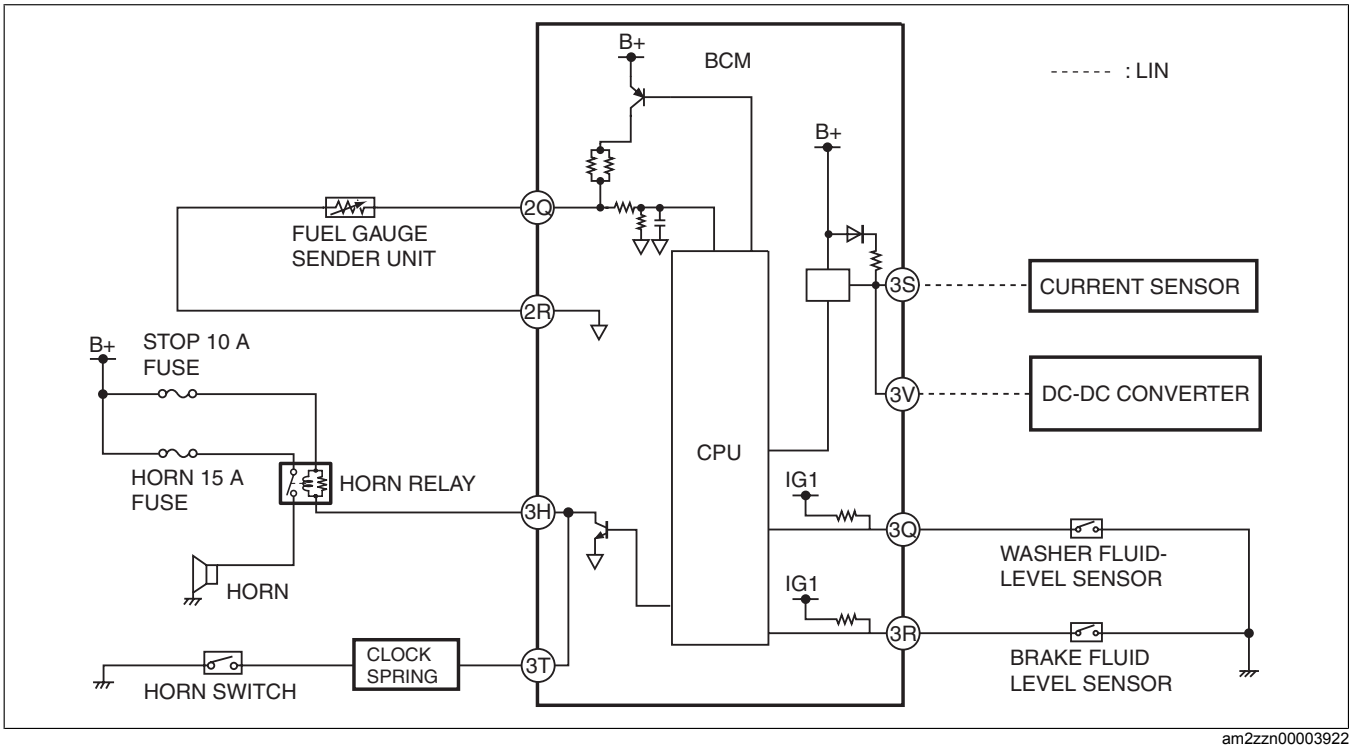
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Door lock, liftgate opener



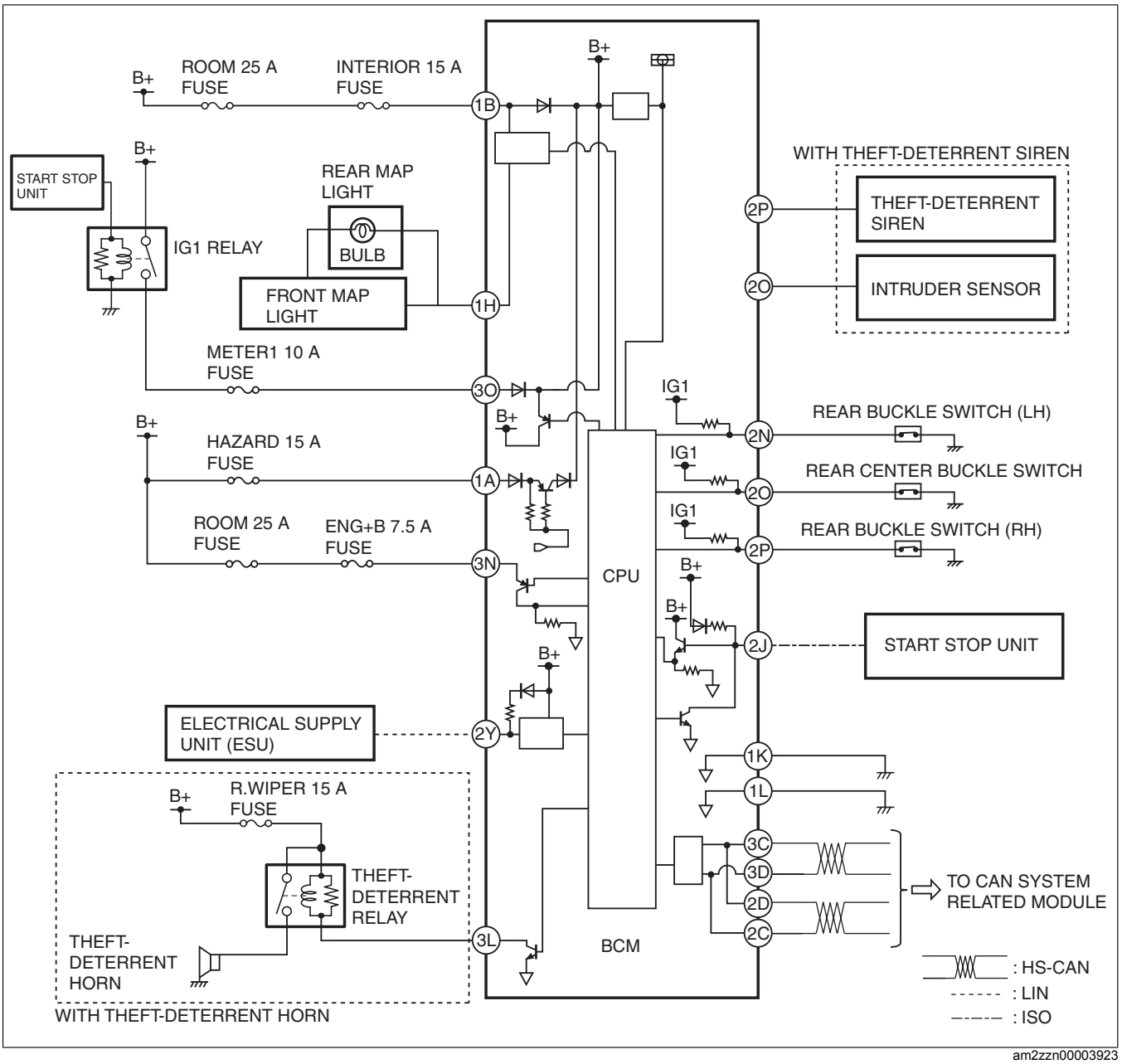
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Fuel gauge sender unit, horn, brake fluid level sensor, current sensor, washer fluid-level sensor



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Power supply, LIN communication, ISO communication, CAN system, rear buckle switch, theft-deterrent system



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Operation

For the operation of each control, refer to the following system operations.

- Front fog light control (See FRONT FOG LIGHT.)
- Turn light system control (See TURN LIGHT SYSTEM.)
- Brake light control (See SMART CITY BRAKE SUPPORT (SCBS).)
- Front map light control (See FRONT MAP LIGHT.)
- Rear map light control (See REAR MAP LIGHT.)
- Cargo compartment light control (See CARGO COMPARTMENT LIGHT.)
- Coming home light system control (See COMING HOME LIGHT SYSTEM.)
- Leaving home light system control (See LEAVING HOME LIGHT SYSTEM.)
- Power door lock system control (See POWER DOOR LOCK SYSTEM.)
- Liftgate opener system control (See LIFTGATE OPENER SYSTEM.)
- Windshield wiper system control (See WINDSHIELD WIPER/WASHER SYSTEM.)
- Rear wiper system control (See REAR WIPER/WASHER SYSTEM.)
- Blower system control (See BLOWER SYSTEM.)
- Rear window defogger control (See REAR WINDOW DEFOGGER/OUTER MIRROR HEATER.)
- Horn control (See HORN.)
- Headlight cleaner control (See HEADLIGHT CLEANER SYSTEM.)

- Theft-deterrent system control (See THEFT-DETERRENT SYSTEM.)

Fail-safe
Exterior lighting system
- If any of the following malfunctions occurs, the BCM turns the TNS and headlights LO on, and stops other exterior light control.
  - Malfunction in CPU in BCM detected
  - CAN communication error (cannot receive light switch position signal)

Auto-light system
- If any of the following malfunctions occurs, the BCM operates using the controls prior to the auto-light malfunction detection.
  - Malfunction in CPU in BCM detected
  - CAN communication error (cannot receive wiper switch position signal)
  - LIN communication error (cannot receive signal from auto-light sensor)

Running light system
- If the BCM detects a vehicle speed signal error, it controls the running light system as a **3 km/h or more** vehicle speed condition.

Interior light control system
- If the following malfunction occurs, the BCM performs control for a continuous **5 s** period using controls prior to the interior light malfunction detection, and after **5 s** have elapsed, it turns on using the initial value.
  - CAN communication error (cannot receive ignition switch condition signal)

Power door lock system
- Not applicable

Liftgate opener system
- Not applicable

Windshield wiper system
- If any of the following malfunctions occurs, the BCM operates using the controls prior to the windshield wiper malfunction detection.
  - Malfunction in CPU in BCM detected
  - CAN communication error (cannot receive wiper switch position signal)
- If an auto stop switch malfunction (wiper operation status unknown) during the wiper operation is detected, the BCM stops the wiper operation.

Front washer system
- If any one of the following malfunctions occurs, the BCM stops control of the front washer.
  - Malfunction in CPU in BCM detected
  - CAN communication error (cannot receive wiper switch position signal)

Auto wiper system
- If any of the following malfunctions occurs, the BCM operates using the controls prior to the auto-wiper malfunction detection.
  - Malfunction in CPU in BCM detected
  - CAN communication error (cannot receive wiper switch position signal)
  - LIN communication error (cannot receive signal from lane sensor)

Blower system
- If the ignition switch condition signal cannot be received or a malfunction signal is received which was sent via CAN communication from the instrument cluster, the following controls are performed.
  - BCM power supply condition is on: Blower relay is turned on
  - BCM power supply condition is off: Blower relay is turned off

Rear window defogger
- Not applicable

Horn
- The BCM turns off the horn relay when a CAN communication error is detected.